

SMART ECOBIN For Safer Environment

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The Smart EcoBin is an intelligent waste management system designed to promote environmental sustainability and improve the efficiency of waste collection. Equipped with multiple sensors, the system monitors both the condition of the bin and its surrounding environment.

An ultrasonic sensor measures the waste level inside the EcoBin, indicating whether the bin is empty, partially filled, or full. This enables timely collection and prevents overflow. A DHT11 sensor records temperature and humidity, ensuring that conditions inside and around the bin are monitored for improved hygiene and maintenance. An MQ135 air quality sensor detects harmful gases and evaluates overall air quality both within the bin and in its surrounding area.

By integrating these sensing technologies, the Smart EcoBin strengthens waste management while supporting a cleaner, healthier, and more sustainable environment. The prototype, along with the dustbin, is shown in Fig. 1, and the required components are listed in the Bill of Materials table.

Circuit and working

Fig. 2 shows the circuit diagram of the Smart EcoBin for environmental applications. It is built around Arduino Nano (MOD1), MQ135 air quality sensor (S1), DHT11 temperature and humidity sensor (S2), ultrasonic sensor HC-SR04 (S3), LCD I2C module, breadboard, Arduino Nano cable, and the EcoBin unit.

The MQ135 sensor is connected to the A1 analogue pin of the Arduino Nano. The DHT11 sensor,



Fig. 1: Author's prototype

toxic, an alarm is triggered to alert nearby individuals. Similarly, in the event of a fire near the bin, the system generates a safety alarm. The ultrasonic sensor detects the level of waste inside the EcoBin. When the EcoBin becomes full, the system generates an alert to notify the owner to dispose of waste at the appropriate collection site.

Thus, the

EcoBin not only manages waste levels but also provides safety and environmental monitoring through integrated sensor-based alerts. One of the key components, the MQ135 gas sensor, detects a wide range of gases harmful to human health. Commonly employed in air quality monitoring systems, it is valued for reliability and versatility. It can sense gases such as ammonia (NH₃), carbon monoxide (CO), alcohol, benzene, smoke, and nitrogen oxides (NO_x), making it effective for monitoring pollution and hazardous conditions.

Software

The Arduino Nano is a compact, versatile microcontroller board based on the ATmega328P. It provides 14 digital input/output pins, 8 analogue input pins, a 16MHz crystal oscillator, a USB connection, a power jack, and a reset button. Its small size makes it

BILL OF MATERIALS

Components	Quantity
Arduino Nano (MOD1)	1
LCD I2C (MOD2)	1
MQ135 module (S1)	1
Buzzer (B1)	1
DHT11 temperature and humidity sensor (S2)	1
Ultrasonic sensor HC-SR04 (S3)	1
Jumper wires	As required
Breadboard	1
USB cable	1
5V power adaptor	1

connected to the A0 pin, measures both temperature and humidity levels. The LCD I2C module displays real-time information, including air quality status (Good, Poor, Very Poor, or Toxic), along with temperature and humidity readings. Arduino Nano and sensor pins are connected as shown in Fig. 2.

The air quality sensor continuously monitors the surrounding air. If quality drops to very poor or